Hackathon Ideas

NOTE

*We nodigen docenten, schoolleiders, studenten, leerlingen en andere geïnteresseerden uit om in teams van 6-8 personen gedurende drie dagen (8, 9 en 10 april 2026) te werken aan een **concreet onderwijsvraagstuk rondom feedback en AI.** (https://www.utwente.nl/onderwijs/pre-university/pro-u/activiteiten/hackathon/)*

Helpen van GenZ studenten met 'gewone' dingen:

https://ukrant.nl/magazine/mij-niet-bellen-de-telefoonangst-van-gen-z/

https://www.tudelft.nl/stories/articles/slimme-chatbots-als-goede-raadgever-voor-angst

https://edition.cnn.com/2026/03/07/health/gen-z-ai-conversations-wellness

- Bellen: huisarts of een docent 'Vooral bellen met iemand die belangrijker is dan ik, vind ik verschrikkelijk.'
- Als ze zo'n telefoontje moet plegen, bereidt ze altijd een script voor in haar hoofd. 'Doe ik dat niet, dan ga ik hakkelen en dan begrijpt de ander me niet.'
- Schrijven van e-mail
- Netwerken, gesprekken starten …
-

MAAR OOK TOETSANGST

Een franse les examen tutor. Iets dat de broer van Japser had gemaakt?

- All kinds of things with avatars (Eriks idea:

Ik ben erg enthousiast geworden over dit project: https://github.com/met4citizen/TalkingHead met een aantal leuke toepassingsvoorbeelden ook op deze pagina. De MotionEngine plugin waarbij je bewegingen kunt besturen via LLM responses lijkt me helemaal leuk om verder uit te testen: lhupyn/motion-engine: Motion engine plugin for TalkingHead 3D avatars
👍 - Teije

- Learning to ask open ended questions: https://edubots.vu.nl/agents/69c6ae6cdfee9d5896cd5aa9/chat/public?s=FktfeL_gzpfKGOVXByIrcnCHcgG0Ez67K9PrccWYVLQ&k=oE-JbP3yFZOJ3AHhdhVrJ3BeECAGhDH6rlWvBcrTLLI
- YourVUture complete digital
- Cogniti Super Apps
- VU Didactic Tips Engager
- Critical Thinking Guide - chuck in a moral axiom or a provocative statement and analyse it:
- Digital Literacy Interactive Learning
- Transactionele Analyse Chatbot
- Your digital Twin: your past you and your future you (something with student well-being, career something …)
- Your Manosphere Escaper
- Your Reflection You
- Your Goal Reminder - set goals, get reminded and reflect
- Your
- Something with a Knowledge Graph (first explain it to me …)
- Your Network Graph ...
- Your AI Literacy Developer
- Your ...
- ….
- … glmph

- Your Digital DeTox Challenger
- Your reflective Diary
- ...
- Your Upper Coach
- Clinical Coach in Cogniti integrating one mini app and two agents
-

1. AI-Powered Peer Feedback Coach

An AI tool that helps students give better peer feedback. Instead of replacing feedback, it coaches students on how to make their feedback more specific, constructive, and actionable, improving both the giver's and receiver's learning.

2. Adaptive Study Choice Navigator

A conversational AI that helps students discover the right study program based on their interests, skills, and motivations, with real-time feedback on their self-reflection quality. Directly fits the hackathon's goal of helping students "sneller op de juiste plek terechtkomen."

3. Canvas Feedback Dashboard with AI Synthesis

An AI layer on top of Canvas that aggregates all feedback a student receives (from teachers, peers, rubrics) into a personal learning dashboard with actionable next steps and trend analysis. Answers the question: "Where am I

improving? Where am I stuck?"

4. AI Teaching Assistant for Formative Assessment

A tool where teachers describe learning goals, and AI generates formative assessment questions mid-lecture, analyzes student responses in real-time, and gives the teacher instant insight into misconceptions, closing the feedback loop faster.

5. Multi-Modal Feedback Translator

AI that takes written teacher feedback and transforms it into the format most useful for each student, visual summaries, action checklists, audio explanations, or simplified language, making feedback more accessible and inclusive.

LITERATURE

Phone call anxiety in students summary

April 8, 2026

Phone-call anxiety in students appears to come less from the phone itself and more from a mix of social anxiety, fear of negative evaluation, low control in real-time interaction, and strong habit with more editable media such as texting. The literature also suggests that lack of practice matters. Students often do better when support combines normalization, structure, and low-stakes rehearsal rather than simple pressure to just call. (Archbell & Coplan, 2021; Brown & Medcalf-Bell, 2022; Kim & Oh, 2023; Rude et al., 2022)

- Justify why this is relevant to feedback in education!
- Open source
- Covid
- Facial/emotion recognition open university 2004 – get feedback on body language à honest mirror
- Research on how people are using chatbots as therapy
- Why would they use our specific chatbot à connect it to canvas? Embed into better prepared tool? In whatsapp?
    - Canvas extrinsic motivation; whatsapp more voluntary, intrinsic motivation
- Make visuals for ads nanobanana jasper might help marketing images
- Put into presentation how we would scale it down to a smaller model? Tailored llm?
- Parallel: big ai/small ai

- More anxious to use tools made by universities because people they know might see it vs chatgpt anonymous unknown
- Divide client/patient and mentor/buddy

Summary table

Theme	Key finding	Main conclusion	Implication
Social anxiety	Social anxiety in university students is linked to less communication with instructors, lower engagement, and worse student experience. (Archbell & Coplan, 2021)	Call avoidance is often one part of a broader communication problem.	Students who fear calls may also avoid help-seeking and contact with staff.
Real-time pressure	Telephone anxiety is associated with heavier use of digital communication and preference for non-call channels. (Kim & Oh, 2023)	Calls feel harder because they require immediate response with less control.	Scripts, practice, and predictable call structure should help.

Uncertainty	Social anxiety predicts phone-related coping through intolerance of uncertainty and anxiety-reduction motives. (Brown & Medcalf-Bell, 2022)	Students are anxious not only about speaking, but about not knowing what will happen in the call.	Support should reduce ambiguity by previewing likely call flow and recovery moves.
Safer substitute media	Anxious young adults often prefer texting because it allows more control over self-presentation, especially in higher-stakes interactions. (Chen & Toma, 2023; Reid & Reid, 2007; Rettie, 2007)	The issue is often channel choice toward safer media, not refusal to communicate at all.	A good bridge is to move from text or email to calls gradually.
Lack of practice	Structured telephone simulation improved pharmacy students' confidence and	Part of the problem is low experience with live professional calling.	Low-stakes repeated practice is one of the clearest supports in the literature.

	self-rated skills. (Rude et al., 2022)		
Prevalence	A study of medical students found 42 percent had at least mild telephobia, though moderate to severe cases were much less common. (Bairwa et al., 2024)	The problem is common enough to matter, but severe impairment is a smaller subgroup.	Broad support can help many students, while some need more targeted anxiety support.
Extra barriers	Telephone anxiety may be stronger when language and cultural barriers are present. (Kim & Oh, 2023)	Anxiety is shaped by context as well as personality.	Extra scaffolding may be needed for non-native speakers and students with less formal call experience.

Key takeaways

- The best-supported causes are:

    - social anxiety
    - fear of negative evaluation
    - low control in live conversation
    - intolerance of uncertainty
    - preference for editable media
    - low practice with formal phone communication (Archbell & Coplan, 2021; Brown & Medcalf-Bell, 2022; Kim & Oh, 2023; Rude et al., 2022)
- The literature supports a skills plus anxiety view more than a simple claim that students have generally lost social skills. (Brown & Medcalf-Bell, 2022; Chen & Toma, 2023; Kim & Oh, 2023)
- The strongest practical implication is that support should combine:
    - normalization
    - structured scripts
    - low-stakes rehearsal
    - gradual exposure
    - added help for students with broader social anxiety or language barriers (Archbell & Coplan, 2021; Kim & Oh, 2023; Rude et al., 2022)

Short literature review

Recent literature suggests that young adults' fear of telephone calls is best understood as a form of communication anxiety shaped by both individual vulnerability and the affordances of the medium. In higher education, social anxiety is linked to reduced academic communication, including less contact with instructors, lower engagement, and poorer student experience (Archbell & Coplan, 2021). This suggests that call avoidance is rarely an isolated behavior. It often sits within a broader pattern of fear of evaluation and withdrawal from high-stakes communication.

A second theme in the literature is that phone calls create a kind of pressure that many students experience as hard to manage. Compared with texting or email, calls require immediate response, offer less time to plan, and reduce control over self-presentation. Recent work on telephone anxiety and media preference shows that heavier reliance on digital communication is associated with greater telephone anxiety, while socially anxious young adults often prefer more editable channels, especially in situations that feel risky or emotionally loaded (Chen & Toma, 2023; Kim & Oh, 2023). Earlier work points in the same direction, finding that anxious users tend to prefer texting over talking because text allows more control and lowers the chance of visible mistakes (Reid & Reid, 2007; Rettie, 2007).

The literature also points to a practical conclusion for student support. Phone-call anxiety appears to be a skills plus anxiety problem, not simply a matter of unwillingness. One recent study found that social anxiety predicted phone-related coping through intolerance of uncertainty and anxiety-reduction motives (Brown & Medcalf-Bell, 2022), which suggests that students may fear not just speaking, but the unpredictability of live interaction itself. At the same time, intervention work with students shows that structured, low-stakes telephone practice can improve confidence and perceived competence (Rude et al., 2022). Taken together, the evidence suggests that effective support should combine normalization, clear call structure, rehearsal, and gradual exposure, with extra attention to students who also face language barriers or broader social anxiety (Kim & Oh, 2023; Rude et al., 2022).

References

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Literature FACT-CHECK by Sofia

Archbell

- Among the results, social anxiety was negatively related to communication with instructors, socio-emotional functioning, and student experiences.

- The academic context appears to be particularly stressful for socially anxious students. Mattick and Clarke (1998) explained that central concerns of individuals with social anxiety include fears of being inarticulate, sounding boring or unintelligent, not knowing how to respond, and being ignored. These fears may become exacerbated in the academic environment, given the participatory and social nature of the classroom à also, networking and building a career (added by me)
- Students report feeling overlooked because of this, it harms their career and studies.

Bairwa

- Fear of taking phone calls = telephobia, considered a part of social anxiety
- Fear of making calls may be associated with concerns about finding an appropriate time to call and fear of being a nuisance. Some sufferers may be anxious about having to "perform" in front of a real or perceived audience at their end of the line.
- At the same time, members of a younger generation who have grown up with digital communication increasingly find both making or receiving phone calls "intrusive," preferring to use media that allow them to "participate in the conversation at the pace they choose."

Brown

- existing literature has indicated a strong link between social anxiety and phone use with increased social anxiety having a strong, positive correlation with mobile phone addiction (due to avoiding social interactions) à this suggests that having a solution for their social anxiety in their phone might be in their safe space, encouraging them to use it; but it can also increase phone dependence, so it needs to be efficient at preparing students to take their skills offline
    - main issue here: uncertainty, making mistakes, fear of negative judgment by accidentally being too loud/rude
    - prefer texting because it's predictable, time to think
    - socially anxious people worry about not only how the counterpart will behave, but also how they themselves will behave in social situations

Chen

- in romantic relationships, in general, people prefer texting over face-to-face when situations are less threatening; breakups and difficult conversations done in person à but the higher the social anxiety, the higher the preference for texting

Kim

- heavy reliance on digital technologies is positively correlated with telephobia
    - millennials more affected than older generations
- issue greater among non-native English speakers

Reid

- anxious people prefer texting, lonely people prefer calling/face-to-face

Rettie (beware, this is from 2007)

- texters enjoy remote social connection
- key characteristics of SMS are its asynchrony and lack of intrusiveness

Rude

- <u>developed and implemented a simulation</u> allowing students to practice interprofessional communication and assess the simulation's impact on students' confidence in providing pharmacy-related interventions to another health care professional <u>via telephone</u>
- significant improvement of <u>confidence</u>
- ability to <u>work independently</u>, communicate an order change to another health care professional, justify recommendations, answer a drug information question, and discuss recommendations in a <u>logical and concise manner</u>
- 5 different case stations simulating various telephone interactions commonly encountered in community pharmacy practice
    - But students were doing the role of pharmacist and student à this is where our AI can offer better feedback than just students

New articles - Sofia

Hawke LD et al. "Digital Conversational Agents for the Mental Health of Treatment-Seeking Youth: A Youth-Engaged Scoping Review." JMIR Mental Health (2025).

- As digital conversational agents expand around the world, a number of ethical issues have been raised. One team set out recommendations for minimal ethical standards in the use of digital conversational agents for youth mental health [17]. Some factors include ensuring confidentiality, being transparent about uses and limits, generating appropriate crisis responses, and protecting against addictive use.

Use of Generative AI for Mental Health Advice Among US Adolescents and Young Adults." JAMA Network Open

- In this cross-sectional study's nationally representative survey, we found that 13.1% of US youths, representing approximately 5.4 million individuals, used generative AI for mental health advice, with higher rates (22.2%) among those 18 years and older. Of these 5.4 million users, 65.5% engaged at least monthly and 92.7% found the advice helpful.
- High use rates likely reflect the low cost, immediacy, and perceived privacy of AI-based advice, particularly for youths unlikely to receive traditional counseling.4 However, engagement with generative AI raises concerns,5 especially for users with intensive clinical needs, given difficulties in establishing and using standardized benchmarks for evaluating AI-generated mental health advice and limited transparency about the datasets training these models.6 Furthermore, Black respondents reported lower perceived helpfulness, signaling potential cultural competency gaps.

https://www.fastcompany.com/91422133/a-third-of-gen-z-has-confided-in-ai-chatbots-chatgpt-therapy-over-humans-mental-health-experts-are-worried

- The generation uses ChatGPT for therapy on everything from work stress to their deepest secrets, and 40% talk to AI for at least an hour every day.
- It listens, it responds thoughtfully, and it never criticizes.
    - "That creates a sense of psychological safety that's often missing in corporate hierarchies. It's about connection, control, and immediacy.

25% of Gen Z describe ChatGPT, Copilot, and other AI bots as their therapist or coach, a friend, or coworker, while 34% admit to confiding in AI chatbots about things they've never told another person. Some 16% say they frequently discuss personal topics such as mental health or relationships with AI, while 33% say they do so occasionall

zijn er al systemen die onze use case behandelen?

Kort antwoord: **er is niemand die precies doet wat jullie doen.** De markt is groot voor algemene angst-apps, maar de specifieke combinatie van gesimuleerd telefoongesprek + bel-angst + exposure-therapie + Nederlandse context is een witte plek. Hier is het landschap:

Wat er wél bestaat

Algemene AI-angstapps (Wysa, Woebot, Youper) Tekstgebaseerde CBT-chatbots. Geen spraak, geen simulatie van echte situaties. Gericht op algemene angst en depressie, niet specifiek op bel-angst.

VR exposure therapy (oVRcome) oVRcome levert klinisch bewezen Virtual Reality Exposure Therapy via smartphone, waarbij gebruikers herhaaldelijk en veilig gevreesde situaties kunnen oefenen. Gericht op fobieën en sociale angst, maar niet specifiek op telefoongesprekken.

Social anxiety apps (FACING IT, Challenger) De Challenger app van Stockholm University nodigt gebruikers uit om steeds uitdagendere sociale interacties aan te gaan, gebaseerd op exposure-therapie principes. Gericht op sociale situaties in het algemeen — geen telefoonsimulatie.

Clare&me — het dichtst in de buurt Clare&me is een AI waarbij je kunt bellen of sms'en voor ondersteuning bij angst en zorgen, gebaseerd op gedragsgezondheidsoefeningen. Maar: Clare belt jou als therapeut, niet als de huisarts of gemeente. Het is geen exposure-oefening voor een specifieke situatie.

AI sales roleplay (Yoodli, Hyperbound) Simuleren gesprekken voor verkooptraining. Zelfde techniek als jullie, maar volledig andere doelgroep en context.

Wat jullie onderscheidt

	Bestaande apps	Jullie aanpak
Medium	Tekst of VR	Echt spraakgesprek
Scenario	Generiek sociaal	Specifiek: huisarts, gemeente, etc.
Doelstelling	Angstvermindering algemeen	Bel-angst specifiek
Methode	Psychoeducatie / chatbot	Graduated exposure via oefening
Taal/context	Engels, internationaal	Nederlands, lokale context
Doelgroep	Breed publiek	Studenten

Wat dit betekent voor jullie

Het is een **echte witte plek** — en dat is zowel een kans als een verantwoordelijkheid. De wetenschappelijke basis (exposure-therapie werkt) is sterk, maar er is nog geen gepubliceerd onderzoek naar AI-gestuurde telefoonsimulatie voor bel-angst specifiek. Dat maakt jullie project potentieel interessant als academische bijdrage.

Is er al een ethische toetsing voorzien voor het onderzoeksproject?

English, ChatGPT: Does a tool like ours already exist?

Yes, parts of this already exist. I did **not** find a clearly established, student-focused product that combines all four layers you described—**mock email feedback, chatbot roleplay for difficult social situations, phone-call simulation, and multi-avatar practice**—into one coherent tool aimed at social/phone-call anxiety. What exists today is a set of adjacent products, usually built for **sales training, leadership training, interview prep, public speaking, language learning, or entertainment roleplay**, rather than for anxious students specifically. (Grammarly)

The closest existing category is **AI roleplay and communication coaching**. Products like **Yoodli**, **Virtual Sapiens**, **Tenor**, **DialogueTrainer**, **PractAIce**, and **CareerTrainer.ai** all center on simulated conversations plus feedback; their common promise is that users can rehearse difficult conversations with AI personas and then receive coaching on what to improve. That means your concept is not category-creating from zero, but your proposed framing—**social-anxiety support for students across written, spoken, phone, and group interactions**—still looks more specific than the mainstream tools I found. (Yoodli)

Here is the deeper read by module.

1) Mock email writing with feedback: this definitely exists in partial form, but mostly as **writing assistance**, not as **scenario-based exposure therapy or communication training**. **Grammarly** offers tone detection and writing assistance; **WriteMail.ai** and similar tools help users improve drafts, adjust tone, and generate replies; **ChatVisor's email tone checker** explicitly positions itself as pointing out whether wording sounds too harsh, vague, or uncertain. What seems less common is a product that turns email into a **practice simulation** with escalating scenarios, rubric-based feedback, and skill tracking for anxious students. (Grammarly)

2) Chatbot practice for networking, saying no, setting boundaries, difficult conversations: this already exists quite strongly in the market, though usually for workplaces rather than students. **Virtual Sapiens** markets AI roleplay coaching for pitches, difficult conversations, and meetings. **PractAIce** says users can choose a scenario, talk to an avatar with its own personality, and then get immediate feedback on tone, structure, word choice, and pace. **CareerTrainer.ai** offers audio-first avatars for difficult conversations. **Tenor** frames the product around roleplay, coaching, and real-world feedback for situations like performance reviews and compensation conversations. These tools show that the roleplay/coaching loop is real and commercially validated. (Virtual Sapiens | Ai Roleplays)

3) Simulation phone calls / voice practice: this also exists, but again not primarily for social-anxiety students. **VoiceGenie** offers an AI phone simulator for mock interviews. **Call Simulator** focuses on immersive conversation simulations for training organizations, with voice realism and measurable confidence gains claimed on its site. **Character.AI** also publicly promotes the ability to "chat, call, text" with AI characters, though that is more general-purpose consumer interaction than a structured coaching product. So voice interaction is available; what seems missing is a clean student-safe layer that explicitly trains people through **phone-call anxiety** scenarios such as calling a landlord, professor, doctor's office, recruiter, or internship contact. (VoiceGenie)

4) Talking to multiple avatars at once: this exists technically, but mostly in **developer platforms** or entertainment-style roleplay rather than polished student training products. **Inworld AI** documents "conversation groups" and multi-character conversation management, including speaker selection and state tracking. **Convai** positions itself for role-play avatars and XR training. **MiocAI** explicitly offers group chats with multiple AI characters. **Character.AI** has "Scenes," but the public material I found points more toward immersive storytelling than robust multi-party communication training. So the multi-avatar piece is feasible and already present in some products, but it is less mature as a mainstream learning UX. (Inworld AI Documentation)

So the answer is:

A tool like this partially exists, but usually in fragments.
I did **not** find a prominent, integrated product that does all of the following together for your target user:

- scenario-based email practice with feedback,
- AI chat rehearsal for networking and boundaries,
- structured phone-call simulation,
- multi-person avatar practice,
- all framed around **student social anxiety / communication confidence**. (Grammarly)

The **core elements** shared by the closest existing tools are fairly consistent:

1. **Scenario-based roleplay**
   Users pick a situation, persona, or goal rather than chatting generically. (Practaice)
2. **AI personas / avatars with behavioral variation**
   The other party has a role, tone, and personality, making practice feel less scripted. (Practaice)
3. **Immediate coaching and feedback**
   Most tools promise feedback on tone, wording, clarity, pacing, or other communication behaviors after the exchange. (Practaice)

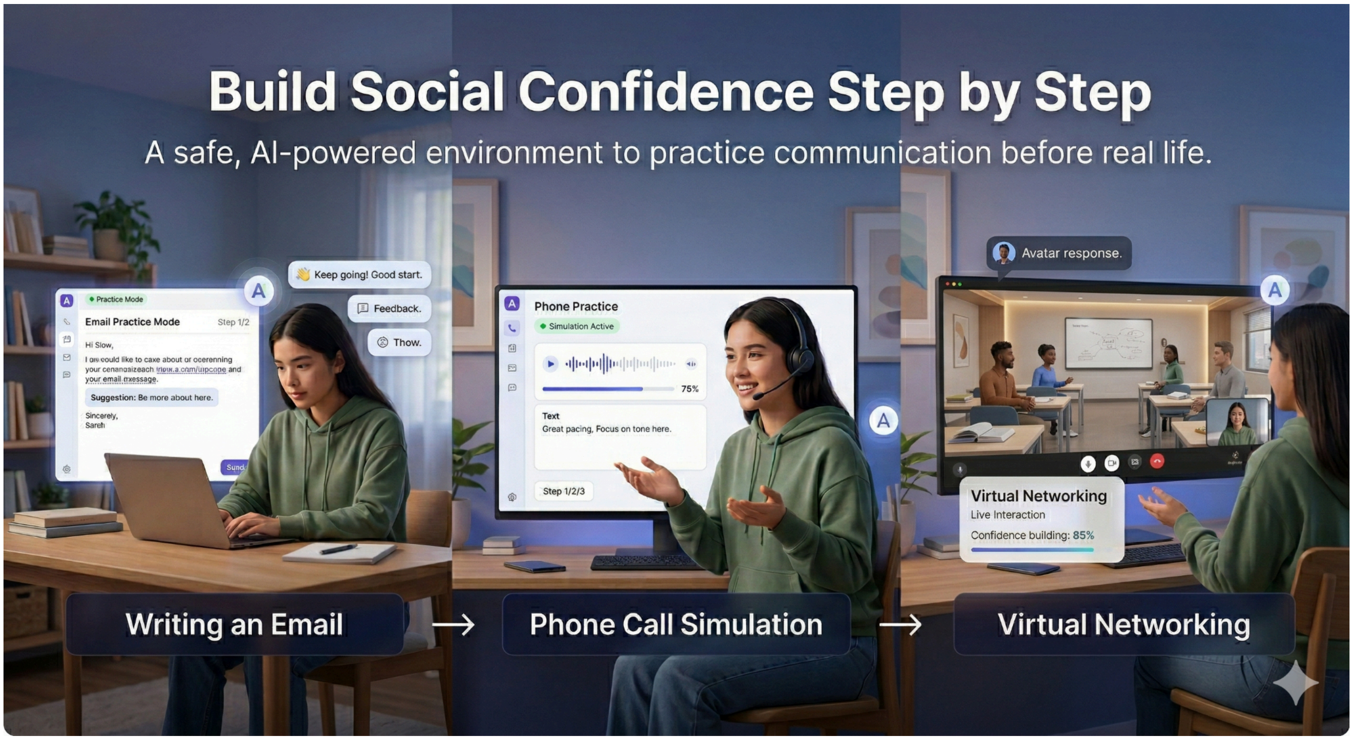
4. **Voice interaction, not just text**
   The stronger products increasingly use voice or audio-first interaction for realism. (Careertrainer.)
5. **Customization and repeatability**
   Users can rerun scenarios, change personas, or practice difficult conversations safely. (Practaice)
6. **Progress measurement**
   Several products position themselves as tracking skill development over time rather than just generating one-off answers. (Practaice)

The gap I see is that existing tools are mostly optimized for **performance coaching**—sales, leadership, interviews, public speaking—not for **graduated social-confidence building** in young people. That gap is probably where your idea becomes distinct. Existing tools tend to ask, “Can this person handle a hard conversation better?” Your concept asks, “Can this student become less avoidant, more capable, and more comfortable across email, one-to-one conversation, phone calls, and group social situations?” That is a different product thesis. (Yoodli)

The main implication is: **you would not be inventing the mechanics, but you may still have a differentiated product if you combine them around a specific user problem and progression model.**

If useful, I can turn this into a competitor map with columns for each of your four modules and show exactly where the market is saturated versus still open.

MARKETING


Build Social Confidence Step by Step
A safe, AI-powered environment to practice communication before real life.
Keep going! Good start.
Feedback.
Practice Mode
Email Practice Mode
Step 1/2
Suggestion: Be more about here.
Phone Practice
Simulation Active
75%
Text
Great pacing, Focus on tone here.
Step 1/2/3
Avatar response.
Virtual Networking
Live Interaction
Confidence building: 85%
Writing an Email
Phone Call Simulation
Virtual Networking